

WEEK 1

Superstore Sales Data Import, Power Query & Data Cleaning

Power BI Desktop — Power Query Editor Workflow Report

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1. Problem Statement

A retail organization wants to analyze its sales data to understand customer purchases, product categories, regional performance, and profitability. Before creating a business intelligence dashboard, the raw Superstore dataset must be imported, cleaned, transformed, and prepared in Power BI. This report documents the Week 1 workflow: importing the dataset, inspecting its attributes, and performing the column and date transformations required to produce a clean, analysis-ready data model.

2. Objectives

- Import data into Power BI Desktop and navigate the Report, Data, and Model views.
- Inspect dataset attributes using the Power Query Editor; verify data types, missing values, and record uniqueness.
- Perform column transformations (rename, split, text trimming, data type coercion).
- Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week).

3. Tasks & Deliverables

- Load the Superstore CSV into the Power Query Editor.
- Standardize text formatting across categorical attributes (Category, Sub-Category, Segment).
- Add custom date columns using Power Query M code / UI transformations.
- Close & Apply transformations to load a clean data model.

4. Environment & Data Source

All work was carried out in Microsoft Power BI Desktop using the built-in Power Query Editor (M engine). The source file was `samplesuperstore.csv`, a comma-delimited export of the public Superstore Sales Dataset containing order-level transaction records — Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, City, and associated Sales, Quantity, Discount, and Profit measures.

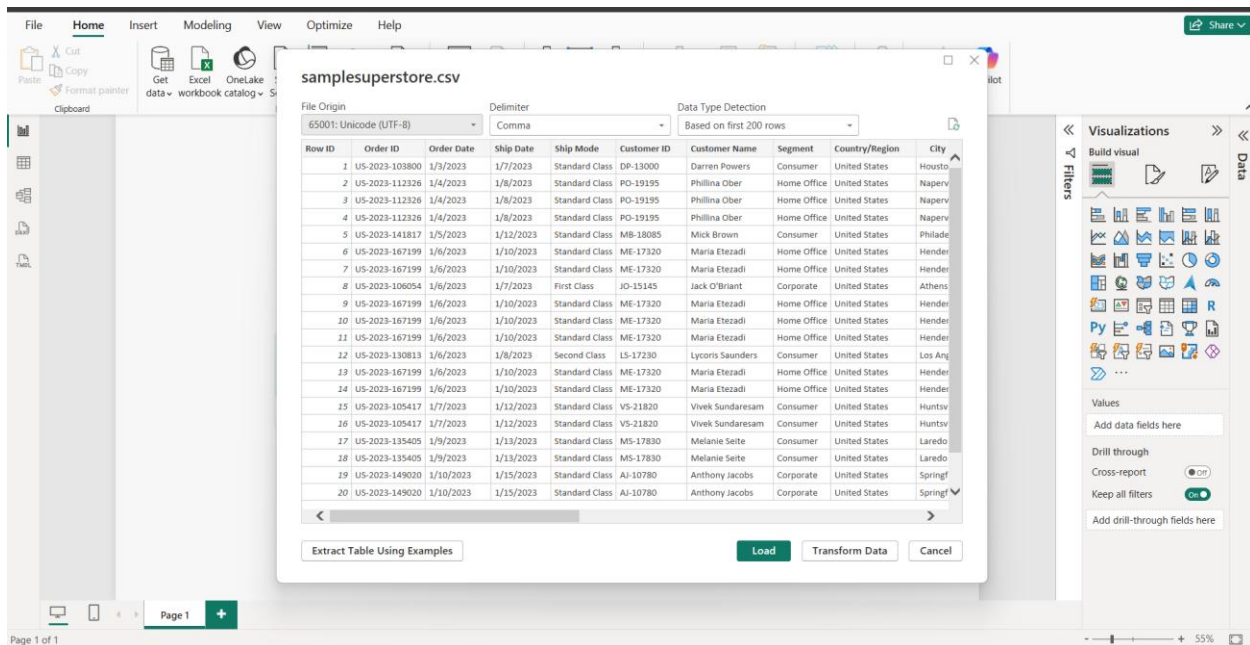
Before any transformation was applied, the dataset was previewed directly in the Get Data dialog to confirm the delimiter, encoding, and header row were detected correctly, avoiding the need to fix parsing errors later in the query.

5. Implementation Walkthrough

The sections below document each step of the Power Query data cleaning process, in the order it was performed, with supporting screenshots taken directly from Power BI Desktop.

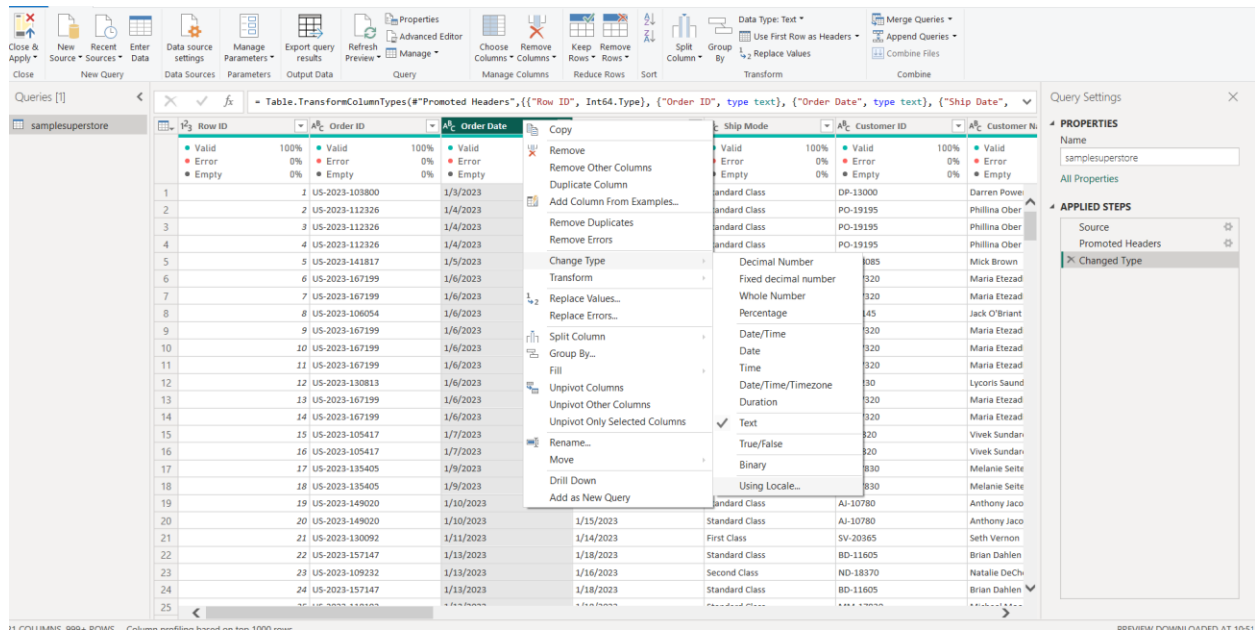
5.1 Importing the Dataset into Power BI

The raw file `samplesuperstore.csv` was imported into Power BI Desktop using Get Data.. The preview confirmed 21+ columns were detected correctly, including Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, and City. Transform Data was selected to open the Power Query Editor rather than loading the raw data directly, so that cleaning could be performed before the model was built.



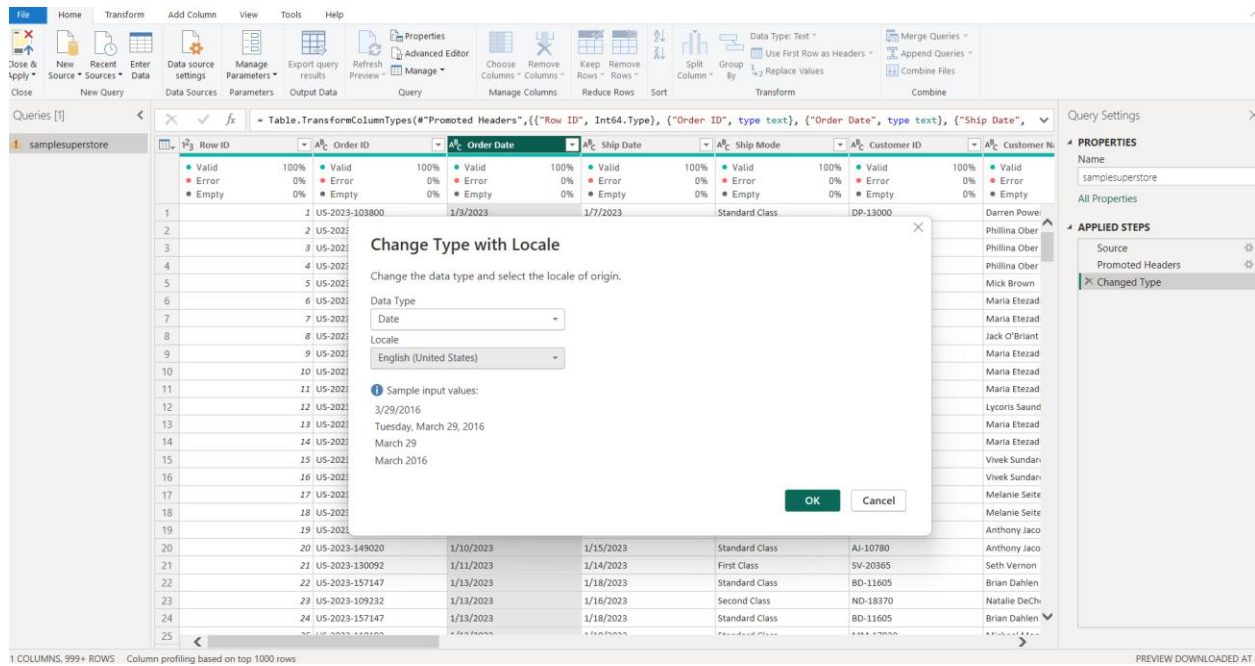
5.2 Inspecting Attributes and Changing Column Data Types

Once inside the Power Query Editor, the Promoted Headers step was applied automatically, and column profiling (based on the top 1,000 rows) was reviewed for each field — checking the Valid / Error / Empty percentages shown under each column header to confirm there were no missing or malformed values. The Order ID and Order Date columns were then reviewed via the Change Type context menu, where Power Query's available type options (Decimal Number, Whole Number, Percentage, Date/Time, Date, Time, Duration, Text, True/False, Binary, and Using Locale) were inspected to choose the correct type for each field.



5.3 Applying Locale-Based Date Type Conversion

Because the Order Date values were stored as text in M/D/YYYY format, a direct Date conversion risked misreading the format. Instead, Change Type → Using Locale was used, which opens the Change Type with Locale dialog. Here the Data Type was set to Date and the Locale was set to English (United States), so Power Query would correctly interpret dates such as 3/29/2016 as March 29, 2016 rather than a European day/month order. The dialog's sample input values were used to confirm the correct interpretation before applying the change.



The same Using Locale technique was repeated for the Ship Date column to ensure both date fields in the dataset were parsed consistently and accurately.

The screenshot shows the Power Query Editor interface. The main data table has columns: Row ID, Order ID, Order Date, Ship Date, Customer ID, and Customer Name. The 'Ship Date' column is highlighted, and a right-click context menu is open. The 'Change Type' option is selected, and a submenu is displayed showing various data types. The 'Using Locale...' option is checked at the bottom of the submenu. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Changed Type with Locale'.

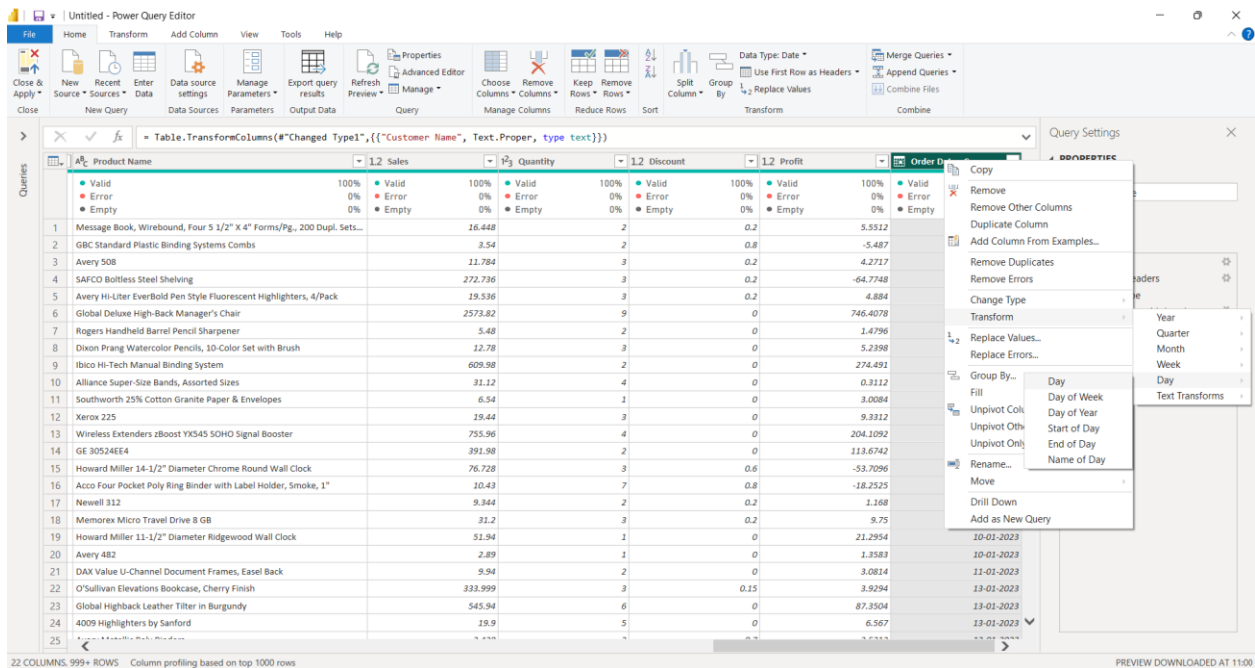
5.4 Duplicating the Order Date Column

To preserve the original, correctly typed Order Date field while still creating a separate set of derived date attributes, the column was duplicated using Duplicate Column from the right-click menu. This produced a new field, Order Date - Copy, on which all subsequent date-part extractions were performed — keeping the base Order Date column untouched for reporting.

The screenshot shows the Power Query Editor interface. The main data table has columns: 1.2 Sales, Quantity, 1.2 Discount, 1.2 Profit, Order Date - Copy, and Order Date. The 'Order Date - Copy' column is highlighted, and a right-click context menu is open. The 'Duplicate Column' option is selected. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Duplicated Column'.

5.5 Extracting Date Components for Time-Based Analysis

With a Date-typed column in place, the Transform ribbon's Date column tools were used to extract temporal attributes needed for later trend analysis. The Transform menu exposes Year, Quarter, Month, Week, and Day sub-menus; under Day, the Day, Day of Week, Day of Year, Start of Day, End of Day, and Name of Day options are available. The Day option was applied to the Order Date - Copy column to extract the numeric day component, producing a new Extracted Day step in the query.



5.6 Standardizing Text Formatting

To keep categorical and name fields consistent for reporting and slicers, text standardization was applied — using Transform → Format → Capitalize Each Word on fields such as Customer Name — recorded in the Applied Steps pane as Capitalized Each Word. This ensures uniform casing across text attributes such as Category, Sub-Category, and Segment before the data is loaded into the model.

5.7 Reordering Columns

After the new derived columns (Order Date - Copy and Extracted Day) were created, Reorder Columns was used to place them in a logical sequence next to their source fields — immediately after Order Date and before Ship Date — so the query output reads in a clear, analysis-friendly column order: Row ID, Order ID, Order Date, Order Date - Copy, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, and so on.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Table.ReorderColumns(#"Extracted Day",{"Row ID", "Order ID", "Order Date", "Order Date - Copy", "Ship Date", "Ship Mode", "Customer ID", "Customer Name",

Order Date	Order Date - Copy	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region
03-01-2023	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Powers	Consumer	United States
04-01-2023	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States
04-01-2023	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States
04-01-2023	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States
05-01-2023	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown	Consumer	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
06-01-2023	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Briant	Corporate	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
06-01-2023	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States
06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
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06-01-2023	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States
07-01-2023	07-01-2023	12-01-2023	Standard Class	V5-21820	Vivek Sundaresam	Consumer	United States
07-01-2023	07-01-2023	12-01-2023	Standard Class	V5-21820	Vivek Sundaresam	Consumer	United States
09-01-2023	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States
09-01-2023	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States
10-01-2023	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States
10-01-2023	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States
11-01-2023	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon	Consumer	United States
13-01-2023	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States
13-01-2023	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie Decherney	Consumer	United States
13-01-2023	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States

22 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name samplesuperstore

APPLIED STEPS

Source

Promoted Headers

Changed Type

Changed Type with Locale

Changed Type with Locale1

Duplicated Column

Changed Type1

Capitalized Each Word

Extracted Day

Reordered Columns

5.8 Closing & Applying the Transformations

With column types corrected, dates parsed by locale, derived date attributes extracted, text standardized, and columns reordered, the final step was to commit the query. Close & Apply was selected from the Home ribbon, which closes the Power Query Editor and loads the cleaned, transformed table into the Power BI data model.

Untitled - Power Query Editor

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Changed Type with Locale1

Duplicated Column

Changed Type1

Capitalized Each Word

Extracted Day

Reordered Columns

6. Applied Steps Summary

The table below summarizes the full sequence of Power Query steps applied to the Superstore dataset during Week 1, as recorded in the Applied Steps pane.

#	Applied Step	Description
1	Source	Loaded samplesuperstore.csv with UTF-8 encoding and comma delimiter.
2	Promoted Headers	Used the first row of the CSV as column headers.
3	Changed Type	Set initial data types (Row ID as Whole Number, Order ID / Order Date / Ship Date as Text).
4	Changed Type with Locale	Converted Order Date to Date type using the English (United States) locale.
5	Changed Type with Locale1	Converted Ship Date to Date type using the English (United States) locale.
6	Duplicated Column	Created Order Date - Copy from Order Date to preserve the original field.
7	Changed Type1	Set the data type on the duplicated date column.
8	Capitalized Each Word	Standardized text casing on the Customer Name field.
9	Extracted Day	Extracted the numeric day-of-month value from Order Date - Copy.
10	Reordered Columns	Repositioned derived columns next to their source fields for readability.

7. Conclusion

By the end of Week 1, the raw Superstore CSV file was successfully imported into Power BI, profiled for data quality, and transformed through a documented sequence of Power Query steps. Date fields were corrected using locale-aware type conversion, a duplicate date column was created to support day-level extraction, text fields were standardized for consistency, and columns were reordered for clarity. Close & Apply loaded the cleaned dataset into the Power BI data model.